

Aperçu — texte à ajouter (surligné = révisé)

Voici le rendu du surlignage fluo jaune. Le fichier **highlighted_edits.tex** contient le code LaTeX exact (avec maths protégées) à copier-coller. Chaque bloc indique où (page/section) le placer.

Abstract (p.1) – 3.2 GS

We eliminate the DPU-hostile Gram-Schmidt operator via a fold-out rewrite followed by a short post-fold fine-tuning that restores accuracy on the DPU-native graph.

§3.1.1 (p.8, after Eq.2) – 1.2/1.3/3.1 fuzzy

Both RAF-DB and FER2013 provide a single label per image; we use no multi-annotator metadata. For a single-label corpus $n_c = \mathbb{1}[c=y]$, so Eq.(2) reduces to $\mu_c = (\mathbb{1}[c=y] + \alpha \cdot \pi_c)/(1+\alpha)$, generated algorithmically, not from annotator votes. We set $\alpha = 0.1$ and π to the empirical class-frequency prior.

§3.1.3 (p.9) – 3.2 GS reframe

Gram-Schmidt is not a no-op at inference: naively removing it breaks the forward path ($0.858 \rightarrow 0.073$). Deployment is a two-step process: fold-out removes the operator, and a short post-fold fine-tuning (10 epochs) recovers accuracy to 0.849.

§4.2.3 (p.20) – 1.1 d-ablation

We ablate $d \in \{4,8,16,32\}$: accuracy stays within a 3-point band with no trend, while cost grows $0.50/1.0/2.05/4.30\times$. We adopt $d=8$ as the accuracy-cost balance.

§4.2.2 (p.17) – 3.4 disgust

Under a matched-backbone comparison, FLEO slightly exceeds the baseline overall (mAP@0.5 0.817 vs 0.812) and improves the hardest minority class — disgust: +4.5 AP ($0.510 \rightarrow 0.554$).

§4.2.3 / Table 7 (p.20) – 3.4 control

As a control, a converged YOLOv8n trained for the same 10 extra epochs gains only +0.2 mAP@0.5 ($0.812 \rightarrow 0.814$), so the folded-graph recovery is genuine re-adaptation, not extra optimization.

§4.4 (p.26) – 2.9 YOLO11/26

We additionally evaluate YOLO11-FLEO ($\text{mAP}@0.5 = 0.802$). YOLO26 could not be integrated. YOLOv8n remains preferred: it is convolution-only and DPU-native, whereas YOLO11's attention is not compilable as a single DPU subgraph.
